ARCHITECTURE JUSTIFICATION

Digital Campus Library Reservation Gateway

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Scenario. Student members search the ISBN catalog, reserve borrowed copies into a FIFO holding queue, receive hold and overdue notices, and track overdue-fine transactions. Head Librarians oversee library operations.

1. Architectural style analysis

STYLE	FIT FOR THIS SCENARIO	TRADE-OFF
Layered	Separates portal, business rules, and data access; straightforward to maintain.	Does not by itself define how multiple user devices share one live queue.
Microservices	Could scale catalog search and notifications independently.	Adds deployment, network, and consistency overhead for a single academic library system.
Client-server	Central server can coordinate member requests, catalog data, FIFO holds, notices, and fine records.	Server availability and capacity need attention as usage grows.

2. Architecture selection & justification

We chose Client-Server Architecture for the Digital Campus Library Reservation Gateway. Member and librarian clients use a shared library server that hosts the application services and repository.

Reason 1 - Consistent FIFO reservations: A central Reservation Manager can assign and update one authoritative queue for each title, avoiding conflicting positions when several members reserve the same borrowed book at once.

Reason 2 - Shared campus access: Student members and Head Librarians can use separate client screens while relying on the same catalog, loan, notification, and fine data.

Security advantage: The server authenticates requests and applies member/librarian roles before exposing member records or changing reservations and fine transactions. Payment details are handled only by the Fine & Payment Service.

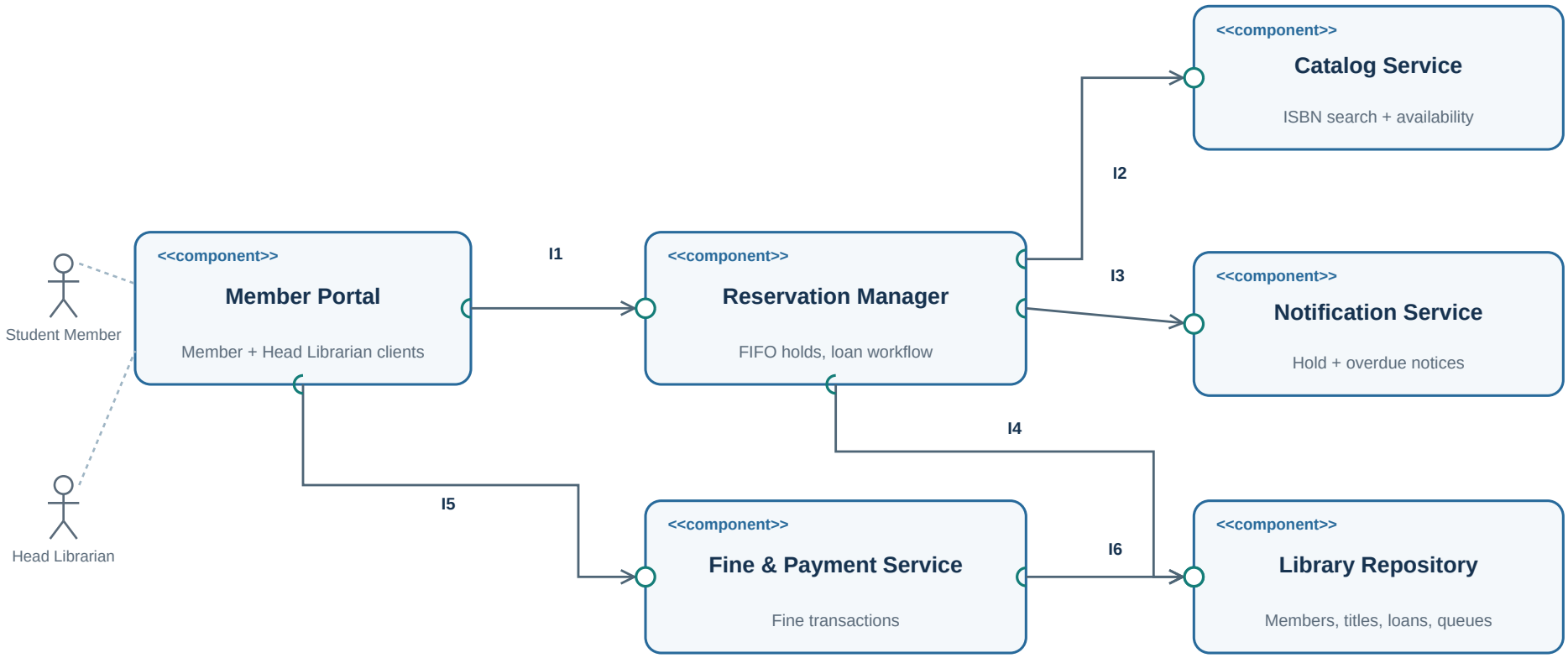
Performance benefit: An indexed ISBN catalog on the central server supports fast, paginated searches and avoids sending the full catalog to each client. Reservation updates stay close to the shared queue data.

COMPONENTS Member Portal | Reservation Manager | Catalog Service | Notification Service
Fine & Payment Service | Library Repository

UML COMPONENT DIAGRAM

Digital Campus Library Reservation Gateway

Client-server architecture | Solid assembly connectors join required sockets to provided interfaces



INTERFACE KEY | IDs ON CONNECTORS MAP TO PROVIDED / REQUIRED SERVICES

I1 IReservation: Portal requires; Reservation Manager provides

I2 ICatalogSearch: Reservation Manager requires; Catalog provides

I3 IHoldNotice: Reservation Manager requires; Notification provides

I4 IReservationData: Reservation Manager requires; Repository provides

I5 IFinePayment: Portal requires; Fine & Payment provides

I6 IFineRecords: Fine & Payment requires; Repository provides

UML notation: open circle = provided interface (ball); socket = required interface; solid line = assembly connector.